Ch 6: Control and Coordination

SECTION A - INTRODUCTION TO CONTROL AND COORDINATION

1 Meaning of Control and Coordination

- All living organisms sense changes in their internal or external environment and respond appropriately.
- Control and coordination are the processes by which an organism regulates and integrates the activites of its organs and systems to bring about a suitable response.

2 Why Multicellular Organisms Need Coordination

- Multicellular organisms are made of many cells, tissues and organs, each with specific functions.
- For smooth working, all parts must act together.
- Coordination helps in:
  - Responding to external changes (e.g., bright light, heat)
  - Maintaining internal balance (homeostasis)
  - Performing daily activities (e.g., movement, digestion)
  - Ensuring survival and reproduction

3 Key Terms in Control and Coordination

Term	Meaning
Stimulus	A change in the internal or external environment that triggers a response.
Receptor	A specialized structure that detects the stimulus.
Coordinator	The control centre (nerve cells or hormones) that processes the information and decides the response.
Effector	An organ or tissue (muscle/gland) that carries out the response.
Response	The action produced by the effector in reaction to the stimulus.

Stimulus → Receptor → Coordinator → Effector → Response

4 Coordination in Animals vs Plants

Feature	Animals	Plants
Control system	Nervous system + Endocrine (hormones)	No nervous system; mainly chemical (hormones)
Speed of response	Very fast (milliseconds)	Relatively slow (minutes to hours)
Mode of signalling	Nerve impulses + Hormones	Plant hormones (e.g., auxin)
Example	Pulling hand away from hot object	Bending of shoot towards light

5 Nervous System – Meaning

- The nervous system is a network of specialized cells (neurons) that detects stimuli and transmits messages to different parts of the body.
- It coordinates quick responses and complex activities.
- In humans, the nervous system has two main parts:
  - **Central Nervous Syatem (CNS):** Brain and spinal cord
  - **Peripheral Nervous System (PNS):** Nerves branching out from CNS to all parts of the body

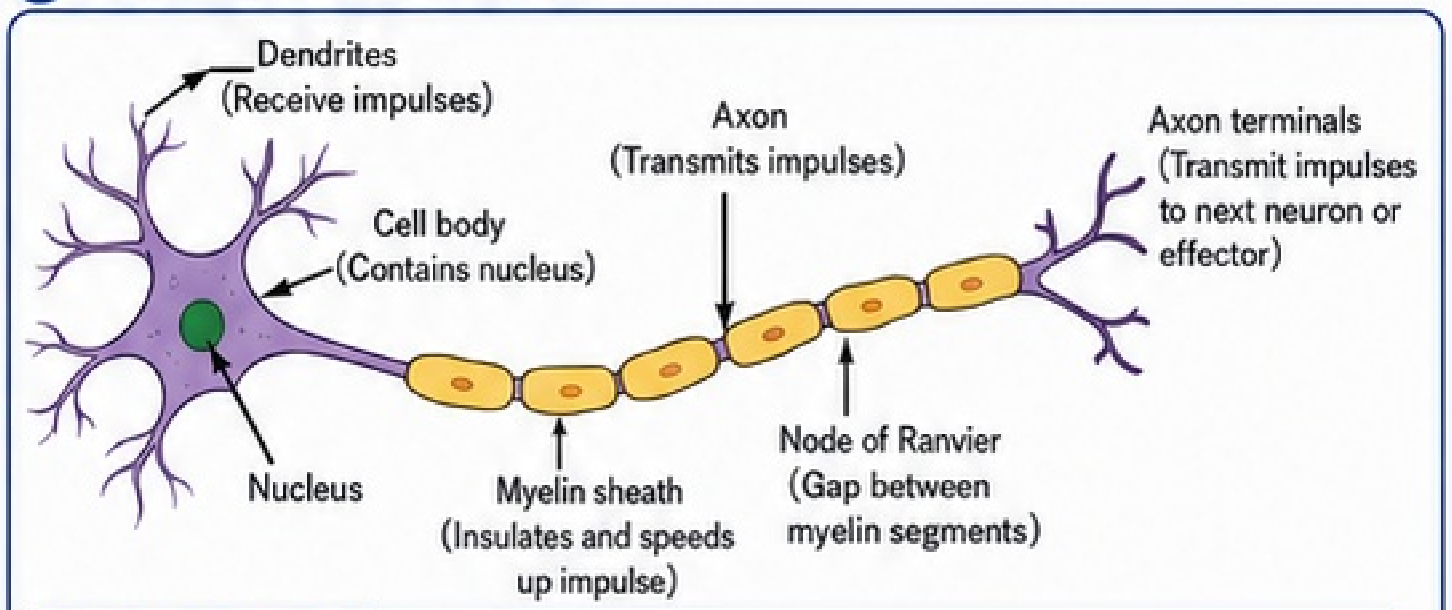
6 Chemical Coordination – Basic Idea

- Besides the nervous system, chemical substances called hormones are released by endocrine glands.
- Hormones are carried by blood to target organs.
- They act slowly but their effects last longer.
- Example: Insulin from pancreas helps in regulating blood sugar.

7 Neuron / Nerve Cell – Meaning

- The basic structural and functional unit of the nervous system is a neuron or nerve cell.
- Neurons are highly specialized cells that receive, process and transmit nerve impulses.

8 Structure of a Neuron



Functions of Parts

- **Dendrites:** Receive impulses from receptors or other neurons.
- **Cell body:** Integrates the impulses and maintains the neuron.
- **Axon:** Carries impulses away from the cell body.
- **Myelin sheath:** Insulates the axon and increases speed of impulse.
- **Node of Ranvier:** Gaps that allow faster conduction (saltatory conduction).
- **Axon terminals:** Pass impulses to next neuron, muscle or gland.

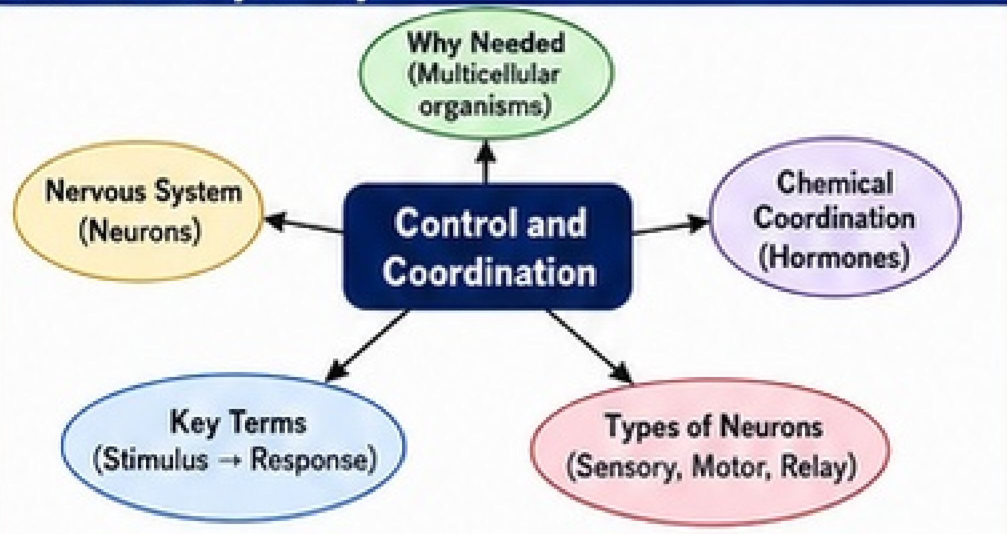
9 Types of Neurons

Type	Function	Example / Location
Sensory (Afferent) Neuron	Carries impulses from receptors to CNS.	From skin, eyes, ears to spinal cord / brain.
Motor (Efferent) Neuron	Carries impulses from CNS to effectors.	From spinal cord / brain to muscles and glands.
Relay (Interneuron)	Connects sensory and motor neurons within CNS.	Found only in brain and spinal cord.

10 Importance of Synapse

- A synapse is the junction between the axon terminal of one neuron and the dendrite or cell body of another neuron.
- Neurotransmitters (chemical messengers) are released at synapse to pass the impulse across the gap.
- Synapse allows one-way transmission of impulses.
- It is essential for communication, learning, memory and coordination of body functions.

11 Concept Map



Key Definitions

Stimulus: A change in the environment that evokes a response.
Receptor: A structure that detects a stimulus.
Coordinator: Control centre that processes information and decides the response.
Effector: An organ or tissue that brings about the response.
Response: The action produced in reaction to a stimulus.
Neuron: A specialized cell that receives and transmits nerve impulses.
Hormone: A chemical messenger secreted by endocrine glands, carried by blood.

Exam Tip

- ✔ Always follow the pathway: Stimulus → Receptor → Coordinator → Effector → Response.
- ✔ Remember: Neurons = Fast response; Hormones = Slow but long-lasting response.
- ✔ Practice neat, labeled diagrams of neuron.
- ✔ Learn the differences between sensory, motor and relay neurons.

Centum Focus

Master the terms, neuron structure and the flow of information. These are foundation concepts for the entire chapter!

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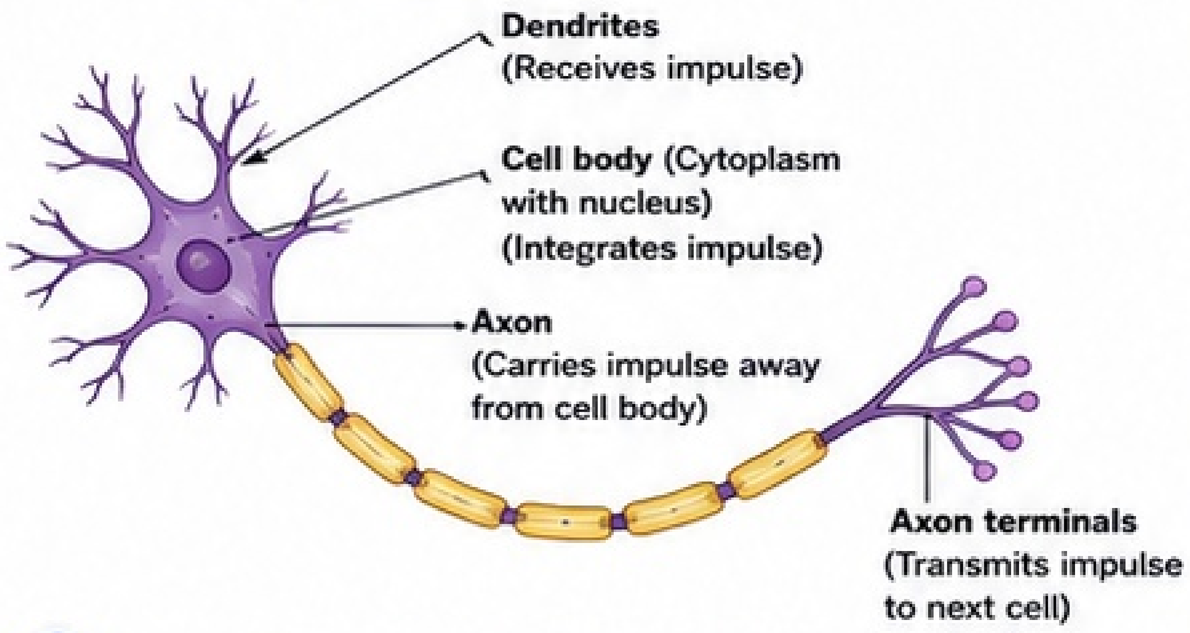
SECTION B - NERVE IMPULSE, REFLEX ACTION AND REFLEX ARC

1 Structure of a Neuron (Nerve Cell)

- Neurons are the structural and functional units of nervous tissue. They transmit nerve impulses.
- A typical neuron has three main parts: dendrites, cell body and axon.



2 How a Nerve Impulse Travels Through a Neuron

1. **Dendrites:** Receive a stimulus (e.g., touch, heat, pain) and start the impulse.
2. **Cell body:** Processes the information.
3. **Axon:** Impulse travels along the axon as an electrical signal.
4. **Axon terminals:** Pass the impulse to the next neuron, muscle or gland through a synapse.

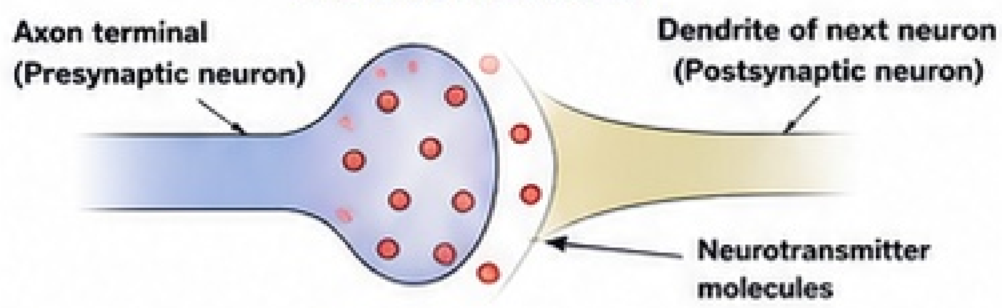
Key Points

- Impulse always moves in one direction: dendrites → cell body → axon → axon terminals.
- Myelin sheath (fatty covering around axon) speeds up the impulse.
- Impulse does NOT travel backward.

3 Synapse and Neurotransmission

- A **synapse** is the junction between the axon terminal of one neuron and the dendrite or cell body of another neuron (or an effector like a muscle/gland).
- There is a tiny gap called **synaptic cleft** between them.
- **Neurotransmitters** are chemical messengers released by the axon terminal.
- Impulses cross the synapse as chemicals, not electricity.

Synapse (Simplified)



1. Arrival of impulse at axon terminal opens vesicles.
2. Neurotransmitters are released into synaptic cleft.
3. They diffuse across and bind to receptors on next neuron.
4. New impulse starts in the next neuron.

Exam Tip

✓ Synapse = Junction
✓ Neurotransmitters = Chemical messengers
✓ Impulse across synapse = Chemical, NOT electrical

4 Stimulus – Response Pathway

Stimulus (Internal or External) → Receptor (Senses change) → Nerve Impulse → Effector (Muscle/Gland) → Response (Action)

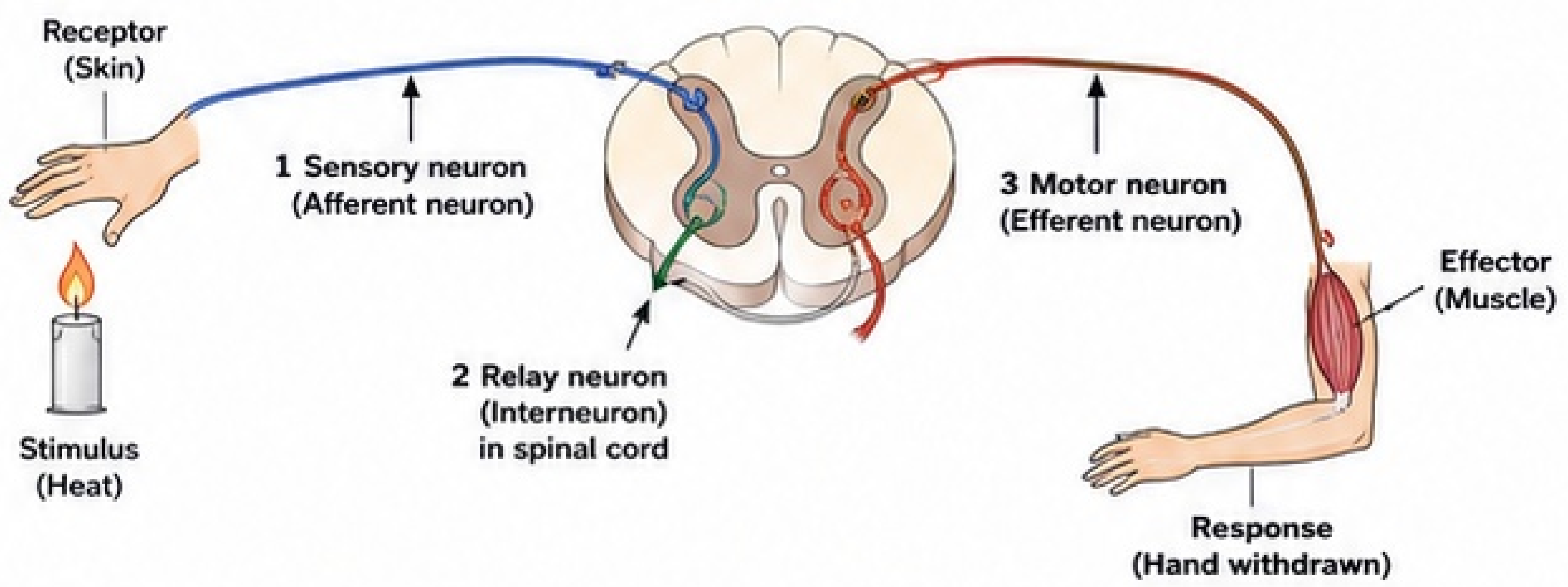
- Receptor detects change and converts it into a nerve impulse.
- Impulse travels to control centre (brain or spinal cord).
- Control centre processes and sends impulse to effector.
- Effector produces response.

5 Reflex Action

- Reflex action is a quick, automatic and involuntary response to a stimulus.
- Purpose: To protect the body from harm and to help in quick survival.
- Examples:
  - Withdrawing hand from a hot object
  - Blinking of eyes when dust enters
  - Knee-jerk when doctor taps below knee
  - Sneezing, coughing, hiccups

6 Reflex Arc – The Pathway of Reflex Action

The complete pathway followed by a nerve impulse during a reflex action.



1 Receptor detects stimulus → 2 Sensory neuron carries impulse to spinal cord → 3 Relay neuron processes and passes impulse to motor neuron → 4 Motor neuron carries impulse to effector → 5 Effector produces response

7 Why Reflex Actions Are Fast?

- Reflex actions do not involve the brain.
- Impulses are processed in the spinal cord.
- Shorter pathway = Less time.
- Saves the body from injury.

Exam Tip

Reflex actions are fast because the impulse travels only up to the spinal cord and immediate response is given.

8 Voluntary Action vs Reflex Action

Feature	Reflex Action	Voluntary Action
Under control of	Spinal cord (automatic)	Brain (conscious)
Needs thinking	No	Yes
Speed	Very fast	Slower
Pathway	Short (reflex arc)	Long (via brain)
Examples	Withdrawing hand, blinking, knee-jerk	Writing, walking, eating, studying

9 Involuntary Actions (Not Reflexes)

- Some actions are automatic but not reflexes.
- Controlled by the brainstem and other parts of the brain.
- Maintain life functions.
- Examples: Breathing, heartbeat, digestion, blood circulation, salivation.

Centum Focus

Remember the KEY WORDS!
Receptor – Detects
Impulses – Travel
Relay – Processes
Effector – Acts
Response – Result

10 Quick Comparison Table

Aspect	Involuntary Actions (Not Reflex)	Reflex Actions	Voluntary Actions
Where controlled	Brainstem / Brain	Spinal cord	Brain (cerebrum)
Purpose	Life sustaining	Protection, quick response	Planned activities
Nature	Automatic	Automatic	Conscious
Examples	Breathing, heartbeat	Coughing, knee-jerk	Writing, speaking
Can we stop?	No	No	Yes

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SECTION C - HUMAN NERVOUS SYSTEM, BRAIN AND SPINAL CORD

1 The Nervous System – Overview

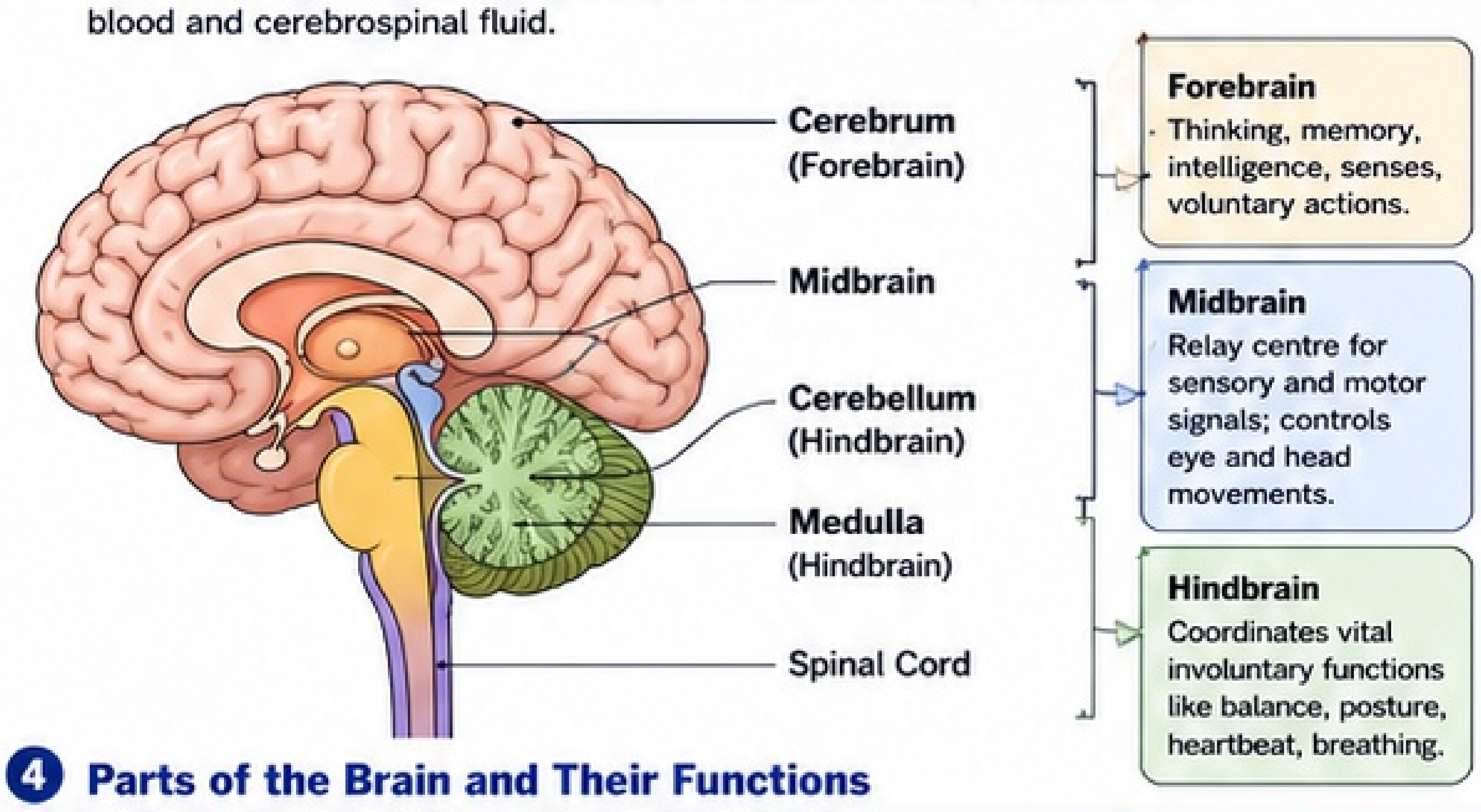
- The nervous system controls and coordinates all activities of the body.
- It receives information from the surroundings and from inside the body, processes it and sends appropriate responses.
- It works with the endocrine system to maintain body functions.

2 Divisions of the Nervous System

- **Nervous System**
  - **Central Nervous System (CNS)** – Consists of brain and spinal cord.
  - **Peripheral Nervous System (PNS)** – Consists of all nerves outside the CNS.
    - **Somatic Nervous System (Voluntary)** – Controls voluntary actions of skeletal muscles.
    - **Autonomic Nervous System (Involuntary)** – Controls involuntary actions of internal organs and glands.
      - **Sympathetic** (Activates body in emergencies)
      - **Parasympathetic** (Relaxes body, conserves energy)

3 The Brain – The Control Centre

The human brain is well protected by the skull. It is richly supplied with blood and cerebrospinal fluid.

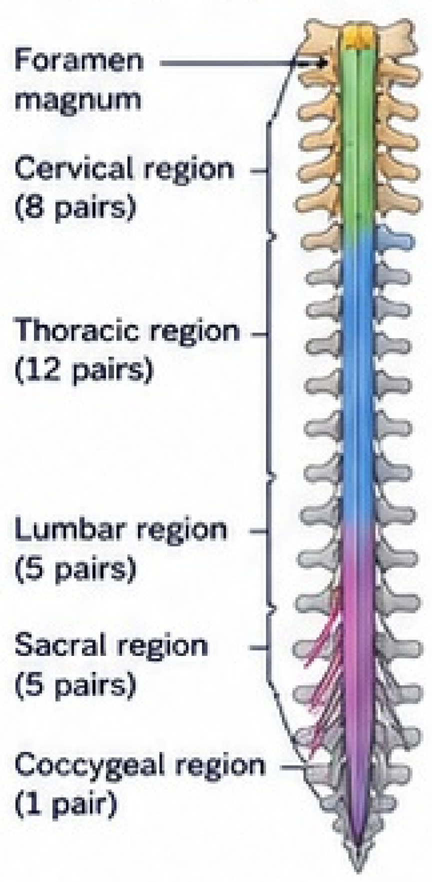


4 Parts of the Brain and Their Functions

Part	Location	Major Functions	Key Roles
Cerebrum (Forebrain)	Upper and largest part	• Thinking, reasoning, problem solving • Memory and learning • Interpretation of sensations (sight, hearing, touch, taste, smell) • Initiates and controls voluntary actions • Language and intelligence	Thinking, memory, learning, decision making, movements of skeletal muscles.
Midbrain	Between forebrain and hindbrain	• Relay station for nerve impulses • Controls reflex movements of eyes and head • Involved in hearing and vision	Eye movements, hearing, visual reflexes.
Cerebellum (Hindbrain)	Behind cerebrum	• Coordinates voluntary muscular movements • Maintains body balance and posture • Controls muscle tone • Ensures smooth, accurate movements	Balance, posture, coordination, smoothness of movements.
Medulla (Hindbrain)	Below cerebellum; continuous with spinal cord	• Controls vital involuntary functions • Regulates heartbeat • Controls breathing rate • Controls blood pressure, swallowing, coughing, sneezing, vomiting	Heartbeat, breathing, blood pressure, digestion reflexes, life-sustaining functions.

5 The Spinal Cord – The Communication Highway

- A long, cylindrical structure running inside the vertebral column.
- Extends from the base of the brain to the lower back.
- Protected by vertebrae and meninges.
- Acts as a pathway for nerve impulses between brain and body.
- Also acts as a reflex centre for many simple and quick actions.



Functions of Spinal Cord

- Conducts impulses:
  - Sensory impulses from body to brain.
  - Motor impulses from brain to body.
- **Reflex centre:** Controls quick involuntary actions without involving the brain.
- Helps in maintaining posture through reflexes.

Examples of Spinal Reflexes:

- Withdrawal of hand from a hot object
- Knee-jerk
- Blinking of eyes
- Sneezing

6 How the Brain Parts Contribute to Key Body Functions

Body Function	Main Part Involved	How It Works
Thinking & Decision Making	Cerebrum	Processes information; forms thoughts; makes decisions.
Memory & Learning	Cerebrum	Stores and recalls information.
Balance & Posture	Cerebellum	Coordinates muscles; maintains balance and upright posture.
Voluntary Movements	Cerebrum & Cerebellum	Cerebrum initiates; cerebellum coordinates smooth execution.
Heartbeat	Medulla	Medulla controls the rate and rhythm of heartbeat.
Breathing	Medulla	Medulla controls breathing rate and depth.
Reflex Actions	Spinal Cord (Reflex Centre)	Responds quickly to stimuli without involving the brain.
Eye & Head Movements	Midbrain	Controls reflex movements of eyes and head.

7 CNS vs PNS – Quick Compare

CNS	PNS
Brain and spinal cord.	Nerves outside CNS.
Control centre.	Communication network.
Processes information.	Carries impulses.
Protected by skull and vertebral column.	Not protected by bones.

8 Pathway of a Reflex (Example)

1. **Stimulus:** You touch a hot object.
2. **Receptor:** Skin detects heat/pain.
3. **Sensory Neuron:** Impulse travels to spinal cord.
4. **Relay Neuron:** In spinal cord (reflex centre).
5. **Motor Neuron:** Impulse sent to effector.
6. **Effector:** Hand muscles contract.
7. **Response:** Hand is quickly withdrawn.

9 Short Recall

- CNS = Brain + Spinal Cord
- PNS = Nerves outside CNS
- Cerebrum = Thinking & memory
- Cerebellum = Balance & coordination
- Medulla = Heartbeat & breathing
- Spinal Cord = Reflexes & pathway for impulses

Exam Tip

- ✓ Always link brain parts to functions for answers.
- ✓ Draw neat labelled diagrams of brain and spinal cord.
- ✓ Give examples of reflexes.
- ✓ In long answers, use tables or flowcharts for clarity.

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SECTION D – ENDOCRINE SYSTEM AND HORMONAL COORDINATION IN ANIMALS

1 Meaning of endocrine glands

- Endocrine glands are special glands that secrete chemical substances called hormones directly into the bloodstream.
- They do not have ducts.

2 Hormones

- Hormones are chemical messengers produced in small amounts by endocrine glands.
- They travel through blood to target organs and control or coordinate body functions.
- They act even in very small quantities.

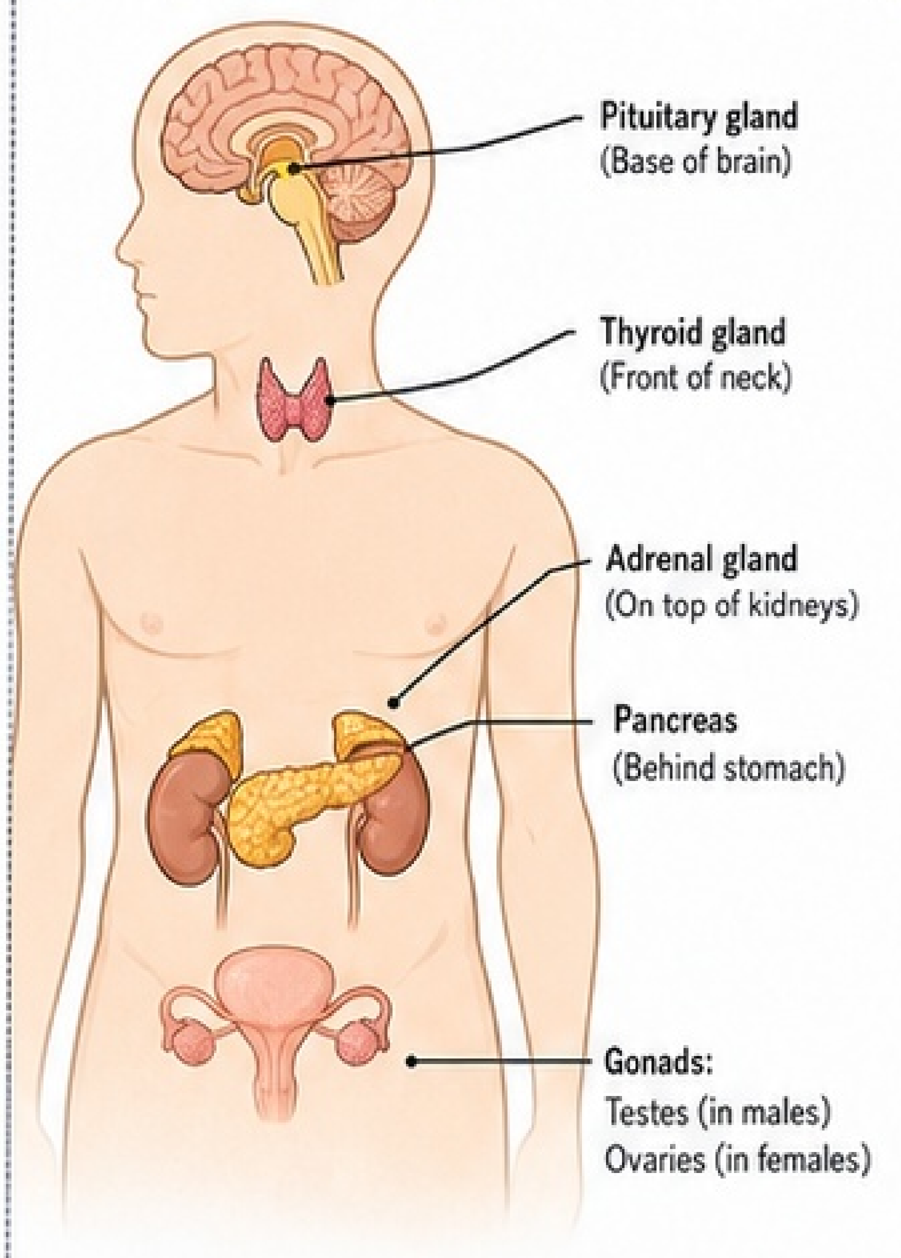
3 Nervous vs Hormonal Coordination

	Nervous Coordination	Hormonal Coordination
1. Mediator	Nerve impulses (electrical signals)	Hormones (chemical substances)
2. Pathway	Along neurons	Through bloodstream
3. Speed	Very fast	Relatively slow
4. Duration	Short-lived	Long-lasting
5. Area of action	Specific and localised	Widespread
6. Example	Reflex action, muscle contraction	Growth, metabolism, reproduction

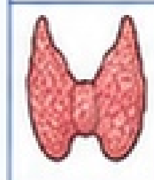
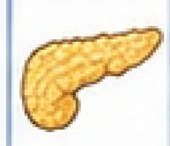
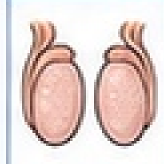
4 Endocrine glands are ductless

- Endocrine glands do not have ducts to carry their secretions.
- Their secretions (hormones) are released directly into blood.
- Examples: Pituitary, Thyroid, Pancreas (islets), Adrenal, Testes, Ovaries.

5 Major endocrine glands in the human body



6 Major endocrine glands – Location & Functions

	**Pituitary gland** (Base of brain)	• Master gland; controls other endocrine glands. • Regulates growth, water balance, reproduction, etc.
	**Thyroid gland** (Front of neck)	• Regulates body metabolism. • Essential for growth and development.
	**Pancreas** (Islets of Langerhans) (Behind stomach)	• Produces insulin and glucagon. • Regulates blood sugar level.
	**Adrenal glands** (On top of kidneys)	• Produce adrenaline. • Helps body respond to stress (fight-or-flight).
	**Testes** (in males) (In scrotum)	• Secrete testosterone. • Responsible for male reproductive functions and secondary sexual characters.
	**Ovaries** (in females) (In pelvis)	• Secrete estrogen and progesterone. • Regulate female reproductive functions and secondary sexual characters.

7 Major hormones at Class 10 level

S. No.	Hormone	Produced By	Main Functions
1.	**Growth hormone** (Somatotropin)	*Pituitary gland*	• Stimulates growth of bones and muscles. • Controls overall growth of the body. • Deficiency in childhood → pituitary dwarfism. Excess in childhood → gigantism.
2.	**Thyroxine**	*Thyroid gland*	• Regulates metabolic rate (carbohydrates, fats, proteins). • Essential for normal growth and brain development. • Deficiency → goitre; in children → mental retardation.
3.	**Insulin**	Pancreas (Islets of Langerhans)	• Lowers blood glucose level. • Helps glucose enter cells and be used for energy. • Deficiency → diabetes mellitus.
4.	**Adrenaline** (Epinephrine)	*Adrenal glands*	• Prepares body for emergency (fight-or-flight). • Increases heart rate, breathing rate, blood pressure. • Mobilises glucose for rapid energy.
5.	**Testosterone**	*Testes*	• Develops male reproductive organs. • Brings about male secondary sexual characters. • Supports sperm formation.
6.	**Estrogen**	*Ovaries*	• Develops female reproductive organs. • Brings about female secondary sexual characters. • Regulates menstrual cycle.

12 Summary Table – Hormone, Gland and Function

Hormone	Gland	Main Function	Deficiency	Excess
Growth hormone	Pituitary	Body growth	Dwarfism	Gigantism
Thyroxine	Thyroid	Regulates metabolism	Goitre, mental retardation	*Weight loss, nervousness*
Insulin	Pancreas	Lowers blood sugar	Diabetes mellitus	Low blood sugar (hypoglycemia)
Adrenaline	Adrenal	*Fight-or-flight response*	*Fatigue, low BP*	*High BP, restlessness*
Testosterone	Testes	*Male characters, sperm formation*	Poor development, infertility	*Aggression, early puberty*
Estrogen	Ovaries	*Female characters, menstrual cycle*	*Irregular periods, infertility*	*Irregular cycles*

8 Why iodine is necessary

- Iodine is essential for the production of thyroxine in the thyroid gland.
- Without iodine, thyroxine cannot be formed properly, affecting growth and metabolism.

9 Goitre (Iodine deficiency disorder)

- Due to lack of iodine in diet.
- Thyroid gland enlarges, causing swelling in the neck (goitre).
- **Symptoms:** swelling in neck, weakness, tiredness, mental dullness.
- **Prevention:** use iodised salt, eat iodine-rich foods (sea fish, seaweed, milk, eggs).

10 Diabetes mellitus (Insulin imbalance)

- Caused by deficiency of insulin or improper action of insulin.
- Blood glucose level rises above normal.
- **Symptoms:** frequent urination, excessive thirst, weight loss, fatigue, blurred vision.
- **Management:** proper diet, *exercise, insulin* injections (if needed).

11 Fight-or-Flight Role of Adrenaline

- *When we face danger or stress, adrenal glands release adrenaline.*
- *It prepares the body to either fight the danger or run away.*
- ***Effects:*** *increases heart rate, breathing rate, blood pressure and energy supply to muscles.*
- ***Example:*** *seeing a snake, exam stress, accident.*

Exam Tips

✓ **Remember:** Glands without ducts → Endocrine glands.
✓ **Learn:** Gland → Hormone → Main function (for diagrams and tables).
✓ Iodine → Thyroxine → Metabolism → Growth.
✓ Insulin → Blood sugar control.
✓ Adrenaline → Emergency response.
✓ Practice drawing the diagram of endocrine glands.

Centum Focus

Focus on understanding how hormones coordinate body functions. Learn the location, hormones and functions of major glands thoroughly.

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SECTION E - COORDINATION IN PLANTS AND PLANT HORMONES

1 Do Plants Show Coordination?

- Plants do not have nerves or muscles.
- Yet they show coordinated responses to internal and external stimuli.
- Responses are slower but well-regulated.
- Coordination in plants is brought about by chemical substances called plant hormones (phytohormones).
- **Examples of coordinated responses:**
  - Shoot bends towards light (phototropism).
  - Roots grow downward (geotropism).
  - Leaves fold in Mimosa on touch (thigmonasty).
  - Stomata close during water stress.
  - Flowers open/close at specific times.

2 Chemical Coordination in Plants

- Information is produced at the site of stimulus.
- It is transmitted by chemical substances (plant hormones) through cells, tissues or vascular tissues (xylem/phloem).
- These hormones act on target tissues to produce a response.
- Hormones are effective in very small amounts.
- Major sites of hormone synthesis: shoot tips, young leaves, developing seeds, roots.

3 Plant Hormones – An Overview

- Plant hormones are natural organic compounds.
- They are synthesized in one part of the plant and transported to another.
- They regulate growth, development and responses to environmental stimuli.
- Five major classical plant hormones:
  Auxin, Gibberellin, Cytokinin, Abscisic Acid (ABA)
  (Ethylene—gaseous hormone—is studied in higher classes).

4 Major Plant Hormones – Comparison

Feature	Auxin (IAA)	Gibberellin (GA)	Cytokinin (CK)	Abscisic Acid (ABA)
Main sites of synthesis	Shoot tips, young leaves, developing seeds	Young leaves, shoot tips, seeds	Root tips (mainly), seeds, developing fruits	Mature leaves, seeds, stems, roots (stress conditions)
Nature	Indole derivative (Indole acetic acid)	Diterpenoid	Adenine derivative (e.g., Zeatin)	Sesquiterpenoid
Primary functions	• Cell elongation • Apical dominance • Tropisms • Root initiation	• Stem elongation • Seed germination • Breaking dormancy • Flowering in some plants	• Cell division • Shoot initiation • Delays senescence • Nutrient mobilization	• Growth inhibitor • Induces dormancy • Closes stomata • Response to stress
Role in growth & development	Promotes elongation of cells in shoots; helps in organ formation.	Promotes stem elongation; bolting; flowering in long-day plants.	Promotes cell division; shoot bud formation; supports overall growth.	Inhibits growth; prevents unnecessary water loss; helps survival.
Role in seed germination	Minor role	Stimulates synthesis of enzymes that mobilize food reserves.	Supports germination along with GA.	Inhibits germination; maintains seed dormancy.
Role in fruit growth	Promotes fruit development; prevents fruit drop (parthenocarpy).	Promotes fruit growth and size in some fruits.	Delays fruit senescence; maintains quality.	Can limit fruit growth during stress.
Role in dormancy & senescence	Delays leaf abscission.	Breaks seed/bud dormancy.	Delays leaf senescence.	Induces seed & bud dormancy; promotes senescence.
Response to stress	Helps in wound healing; root growth in stress.	Helps plants grow under adverse conditions.	Helps maintain growth under moderate stress.	Key stress hormone: triggers stomatal closure, drought response.
Movement in plant	Polar transport (cell to cell)	Through xylem	Through xylem (mainly root → shoot)	Through xylem (phloem in some cases)

5 Hormones in Action – Key Roles in Detail

Auxin (IAA)

- Promotes cell elongation in shoots.
- Causes bending towards light (phototropism).
- Maintains apical dominance (suppresses side buds).
- Promotes adventitious root formation.
- Prevents premature leaf fall and fruit drop.
- Synthetic auxin (NAA) is used in rooting cuttings.

Gibberellin (GA)

- Increases cell division and elongation (tall growth).
- Breaks seed dormancy; stimulates germination.
- Promotes flowering in some long-day plants.
- Increases fruit size (e.g., grapes).
- Overcoming rosette habit in some plants.

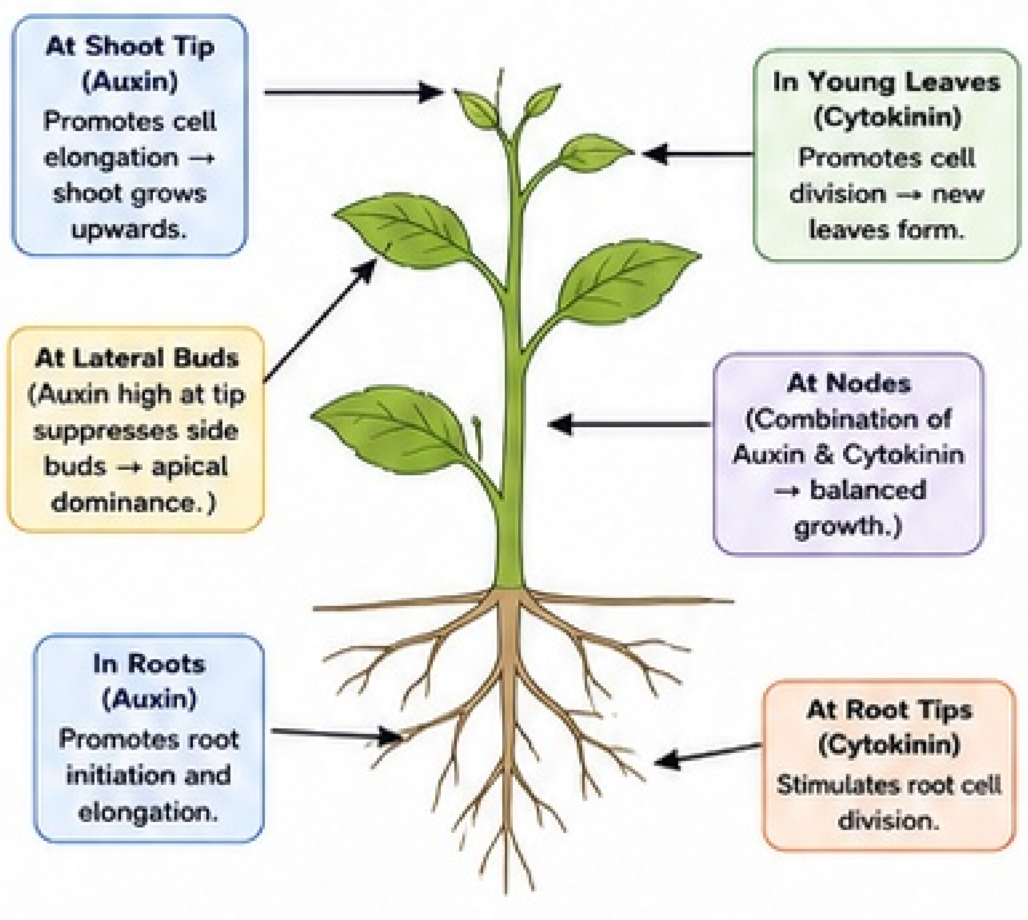
Cytokinin (CK)

- Promotes cell division (cytokinesis).
- Stimulates shoot bud formation (tissue culture).
- Delays leaf senescence (keeps leaves green longer).
- Mobilizes nutrients from old leaves to new tissues.
- Balances auxin for proper growth.

Abscisic Acid (ABA)

- Inhibits growth (antagonistic to auxin and GA).
- Induces seed and bud dormancy.
- Causes stomatal closure → reduces water loss.
- Helps plants survive drought and cold.
- Accumulates in leaves during water stress.

6 Hormone Action in a Plant (Shoot & Root)



7 Growth, Dormancy, Wilting & Stress

- **Growth:** Auxin, Gibberellin and Cytokinin promote growth by elongation or division of cells.
- **Seed Germination:** Gibberellin stimulates; ABA inhibits.
- **Fruit Growth:** Auxin and Gibberellin increase size; Cytokinin delays ripening.
- **Dormancy:** ABA induces dormancy in seeds and buds; GA breaks it.
- **Wilting (temporary):** High ABA → stomata close → water loss reduced → wilting reduces.
- **Responee to Stress:** ABA is the key hormone. It helps plants survive drought, cold, salinity by reducing water loss and metabolic activity.

8 Key Definitions

- **Plant Hormones:** Chemical messengers formed in minute amounts that regulate growth and responses in plants.
- **Auxin (IAA):** Hormone that promotes cell elongation, root initiation and tropic responses.
- **Gibberellin (GA):** Hormone that promotes stem elongation, seed germination and breaking dormancy.
- **Cytokinin (CK):** Hormone that promotes cell division and shoot initiation.
- **Abscisic Acid (ABA):** Hormone that inhibits growth, induces dormancy and helps in stress tolerance.

9 Integrated Concept

Plant growth is the result of the balanced action of hormones:

Growth Promoters: Auxin, Gibberellin, Cytokinin ⟷ **Growth Inhibitor**: Abscisic Acid (ABA)

Balanced action → Normal growth
Imbalance → Abnormal growth / Stress

10 Memory Table – One Look Revision

Auxin	Elongation, Apical dominance, Roots, Phototropism, Prevents fruit drop
Gibberellin	Stem elongation, Germination, Flowering, Fruit growth, Breaks dormancy
Cytokinin	Cell division, Shoot formation, Delays senescence, Nutrient mobilization
ABA	Inhibits growth, Dormancy, Stomatal closure, Stress survival

11 Concept Map

Plant Hormones
- **Auxin**: Elongation, Tropisms, Rooting
- **Gibberellin**: Growth, Germination, Flowering
- **Cytokinin**: Cell division, Shoot growth, Delays aging
- **ABA**: Dormancy, Stomatal closure, Stress tolerance

Exam Tip

✓ Memorize sites of synthesis and key functions of all four hormones.
✓ Always relate hormone to its role in growth, germination, dormancy and stress.
✓ Use examples: phototropism (Auxin), germination (GA), tissue culture (CK), stomatal closure (ABA).

Centum Focus

Practice a table comparing plant hormones and draw hormone action diagram in exam.

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SECTION F - TROPIC MOVEMENTS AND NASTIC MOVEMENTS IN PLANTS

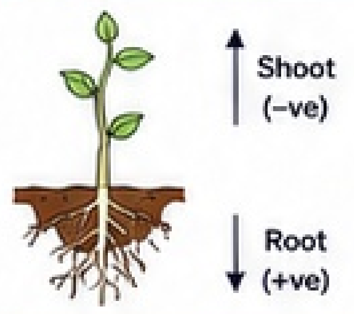
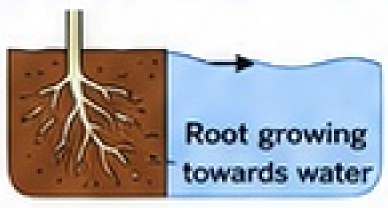
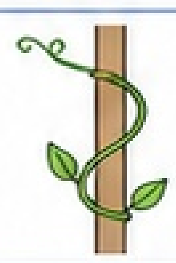
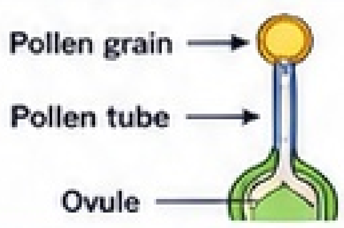
1 Movement in response to stimuli

- Plants, though fixed in position, show movements in response to external stimuli.
- A **stimulus** is any change or signal in the environment that triggers a response.
- Stimuli for plants include light, gravity, water, touch, chemicals, temperature, etc.
- These movements help plants grow in a favourable direction and survive.

2 Tropism – Definition

Tropism: The growth movement of a plant part in response to a stimulus, in a particular direction. It is a directional growth movement.

3 Types of Tropic Movements (with examples)

(i) **Phototropism** (Response to light)	Growth movement in response to light. **Example:** Shoot of a plant grows towards light.	
(ii) **Geotropism / Gravitropism** (Response to gravity)	Growth movement in response to gravity. **Example:** • Roots grow towards gravity (positive geotropism). • Shoots grow away from gravity (negative geotropism).	
(iii) **Hydrotropism** (Response to water)	Growth movement in response to water. **Example:** Roots grow towards water source.	
(iv) **Thigmotropism** (Response to touch)	Growth movement in response to touch. **Example:** Tendrils of pea plant coil around a support when touched.	
(v) **Chemotropism** (Response to chemicals)	Growth movement in response to chemicals. **Example:** Pollen tube grows towards the ovule due to chemical attractants.	

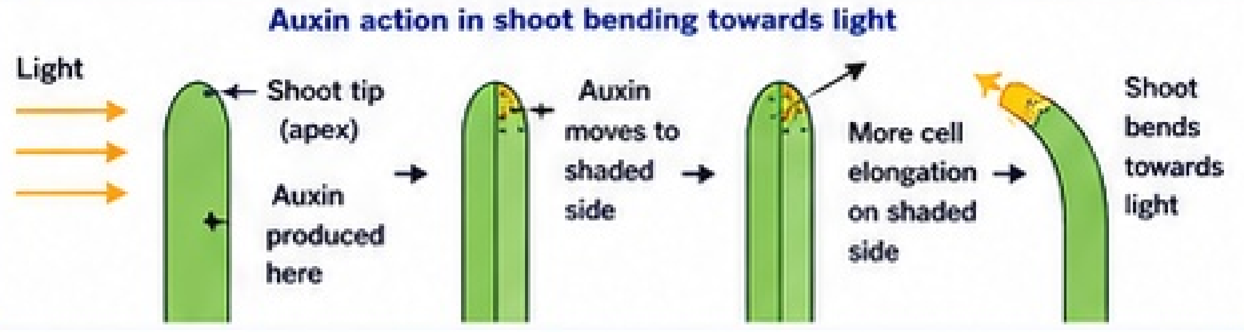
4 Positive and Negative Tropism

- **Positive tropism:** Growth towards the stimulus. Example: Root towards gravity (positive geotropism).
- **Negative tropism:** Growth away from the stimulus. Example: Shoot away from gravity (negative geotropism).

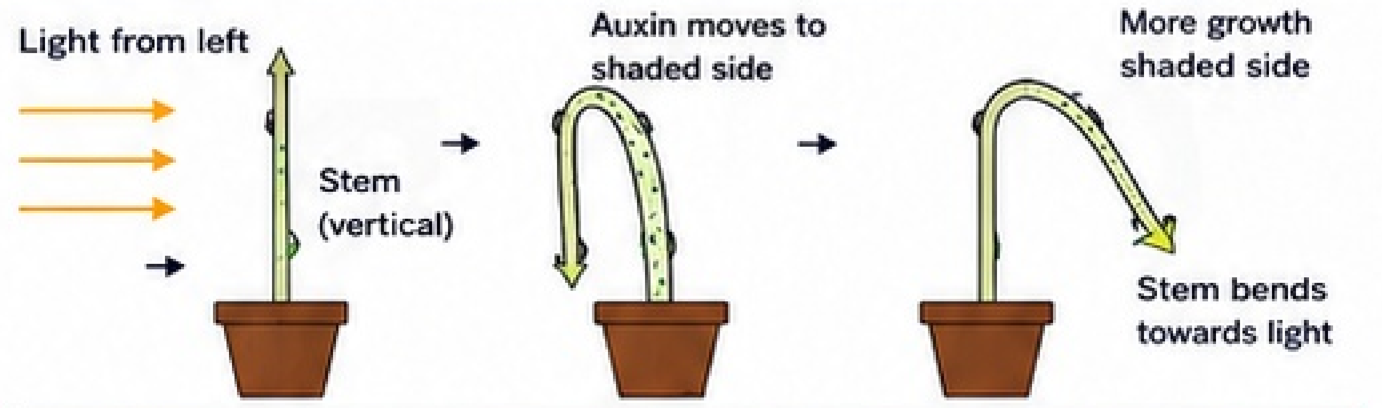
5 How Auxin Causes Bending in Shoots (Phototropism)

1. Light coming from one side of the plant is detected at the tip (apex) of the shoot.
2. Auxin (a growth hormone made in the shoot tip) redistributes to the shaded side.
3. In shoots, auxin stimulates cell elongation.
4. Cells on the shaded side elongate more than those on the lit side.
5. The shoot bends towards the light.

Auxin action in shoot bending towards light



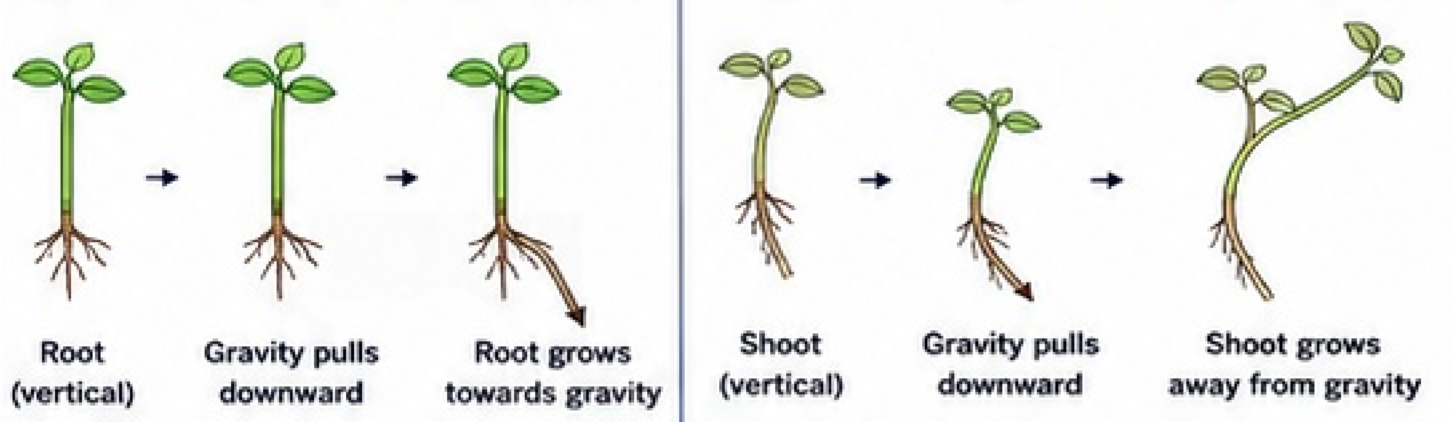
6 Phototropism (in detail)



Explanation: Auxin accumulates on the shaded side of the shoot → cells there elongate more → shoot bends towards light.

7 Geotropism / Gravitropism (in detail)

(a) Root – Positive Geotropism (+ve) | **(b) Shoot – Negative Geotropism (–ve)**



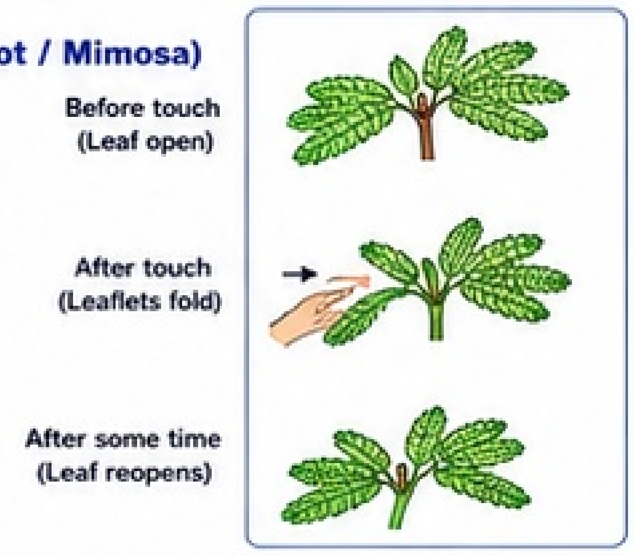
Explanation: Roots show +ve geotropism (towards gravity) and shoots show –ve geotropism (away from gravity).

8 Nastic Movement – Definition

Nastic Movement: Non-directional movement of a plant part in response to a stimulus. The movement is independent of the direction of the stimulus and is usually due to changes in turgor pressure in cells.

9 Nastic Movement – Example (Touch-me-not / Mimosa)

- Leaves of Mimosa *pudica* (touch-me-not plant) fold up and droop when touched.
- It is a nastic movement called **thigmonasty** (response to touch).
- **Cause:** Touch causes loss of water (decrease in turgor pressure) in cells at the base of leaflets → cells become flaccid → leaflets fold.
- **Function:** Protects the plant from injury and reduces water loss.



10 Tropic vs Nastic Movement – Comparison

Feature	Tropic Movement	Nastic Movement
1. Nature	Directional growth movement	Non-directional movement (not growth)
2. Dependence on direction of stimulus	Depends on direction	Independent of direction
3. Cause	Differential growth (usually due to auxin)	Change in turgor pressure in cells
4. Speed	Slow	Fast
5. Example	Phototropism, Geotropism, Hydrotropism, Thigmotropism, Chemotropism	Thigmonasty in Mimosa, Folding of leaves in some plants, Flower opening/closing
6. Reversibility	Usually permanent growth change	Usually temporary and reversible

11 Key Takeaways

- ✓ Tropic movements are growth movements towards/away from a stimulus.
- ✓ Plants use tropisms to grow in favourable directions for survival.
- ✓ Auxin plays a key role in directional growth, especially in shoots.
- ✓ Nastic movements are non-directional and usually due to turgor changes.
- ✓ Mimosa shows thigmonasty—a classic example of nastic movement.

Ch 6: Control and Coordination

SECTION G - IMPORTANT COMPARISONS AND HIGH-YIELD CONCEPTS

1 Nervous Coordination vs Hormonal Coordination

Feature	Nervous Coordination	Hormonal Coordination
System	Nervous system	Endocrine (hormonal) system
Messenger	Nerve impulses (electrical signals)	Hormones (chemical substances)
Pathway	Along neurons	Through blood
Speed	Very fast	Slow
Duration	Short-lived	Long-lasting
Specificity	Very specific (to particular muscles/glands)	Less specific (affects many cells with similar receptors)
Control	Voluntary and involuntary	Mostly involuntary
Examples	Reflex actions, movement, secretion of saliva	Growth, metabolism, puberty, blood sugar regulation
Main Function	Immediate response and rapid adjustments	Long-term regulation and maintaining homeostasis

2 Reflex Action vs Voluntary Action

Feature	Reflex Action	Voluntary Action
Definition	Involuntary, automatic and rapid response to a stimulus	Conscious, deliberate action
Control Centre	Spinal cord (without brain involvement)	Brain (cerebrum)
Awareness	Occurs without awareness	Occurs with awareness
Speed	Very fast	Relatively slow
Pathway	Short pathway (receptor → spinal cord → effector)	Long pathway (receptor → brain → effector)
Can be stopped?	No	Yes
Fatigue	Does not cause fatigue	Can cause fatigue
Examples	Withdrawing hand from a hot object, blinking of eyes	Writing, walking, reading

3 Sensory Neuron vs Motor Neuron vs Relay Neuron

Feature	Sensory Neuron (Afferent)	Motor Neuron (Efferent)	Relay Neuron (Interneuron)
Function	Carries impulses from receptor to CNS	Carries impulses from CNS to effector (muscles/glands)	Connects sensory and motor neurons in CNS
Cell Body Location	Dorsal root ganglion (outside spinal cord)	Ventral horn of spinal cord	Within CNS (spinal cord/brain)
Direction of Impulse	Receptor → CNS	CNS → Effector	Sensory neuron → Motor neuron
Example	Pain receptors in skin sending signals	Nerves sending signals to arm muscles	Neurons in spinal cord during reflex action
Myelin	Usually myelinated	Usually myelinated	Usually unmyelinated
Role in Reflex Arc	First neuron	Third neuron	Second neuron (linking neuron)

4 Plant Coordination vs Animal Coordination

Feature	Plant Coordination	Animal Coordination
Control System	No nervous system or endocrine system	Nervous and endocrine systems
Mode	Growth-based responses (hormones like auxins)	Electrical (nervous) and chemical (hormones)
Speed	Slow	Fast
Nature of Response	Directional or non-directional movements	Reflex, instinctive and voluntary actions
Examples	Phototropism, geotropism, nastic movements	Reflex action, walking, heartbeat regulation
Purpose	Adaptation and survival	Adaptation, survival and complex behaviour

5 Endocrine Gland vs Exocrine Gland

Feature	Endocrine Gland	Exocrine Gland
Secretion	Hormones	Enzymes, sweat, mucus, milk, saliva
Ducts	No ducts	Has ducts
Mode of Release	Directly into blood	Through ducts onto body surface or into a cavity
Range of Action	Affects distant target organs	Acts locally
Examples	Pituitary, thyroid, adrenal	Sweat glands, salivary glands, pancreas (exocrine part)

6 Tropic Movement vs Nastic Movement

Feature	Tropic Movement	Nastic Movement
Direction	Directional (towards/ away from stimulus)	Non-directional (independent of direction)
Nature	Growth-based (irreversible)	Non-growth-based (reversible)
Stimulus Type	Unidirectional (e.g., light, gravity)	Any direction (e.g., touch, shock)
Speed	Slow	Fast
Examples	Phototropism, Geotropism	Touch-me-not leaf folding, Dandelion seed closing

7 Daily Life Examples (High-Yield for Exams)

Example 1: Withdrawing hand from hot tea cup (Reflex Action)

When you touch a hot cup of tea, heat receptors in the skin send impulses through sensory neurons to the spinal cord. Relay neurons process the signal and motor neurons send impulses to the arm muscles to pull hand away immediately—before the brain realises.

Type: Reflex Action **Pathway:** Short (Spinal Reflex Arc)

Example 2: Braking a bicycle on seeing an obstacle (Voluntary Action)

Your eyes (receptors) detect the obstacle. The signal goes to the brain which analyses it. You decide to press the brake. The brain sends impulses through motor neurons to the hand and leg muscles.

Type: Voluntary Action **Pathway:** Long (Through Brain)

8 Concept Map: From Stimulus to Response

Stimulus (Change in internal or external environment) → **Receptor** (Detects the stimulus and sends impulse) → **Sensory Neuron** (Carries impulse to CNS (brain/ spinal cord)) → **Relay Neuron** (Processes and relays impulse in CNS) → **Motor Neuron** (Carries impulse from CNS to effector) → **Effector** (Muscle or gland that responds) → **Response** (Action or secretion produced)

Key Points
- This pathway is called the **Path of Nerve Impulse.**
- **Reflex actions** use **spinal cord** as the control centre (no brain involvement).
- **Voluntary actions** involve brain for decision making.
- **Hormones follow a different pathway: Gland → Hormone → Blood → Target Organ → Response.**

Exam Tips

✓ Learn the differences in tables.
✓ Draw neat diagrams of reflex arc.
✓ Understand examples from daily life.
Tables = High marks in short answers!

Centum Focus

Focus on comparing systems and pathways.
Practice flowcharts and labelled diagrams.
Write crisp points in answers.
These are scoring areas in exams!

Quick Recall (One-Liners)

- Nervous coordination is fast; hormonal is slow.
- Reflex actions are automatic; voluntary actions are conscious.
- Auxin helps plants show tropic movements.
- Hormones are chemical messengers; nerve impulses are electrical.

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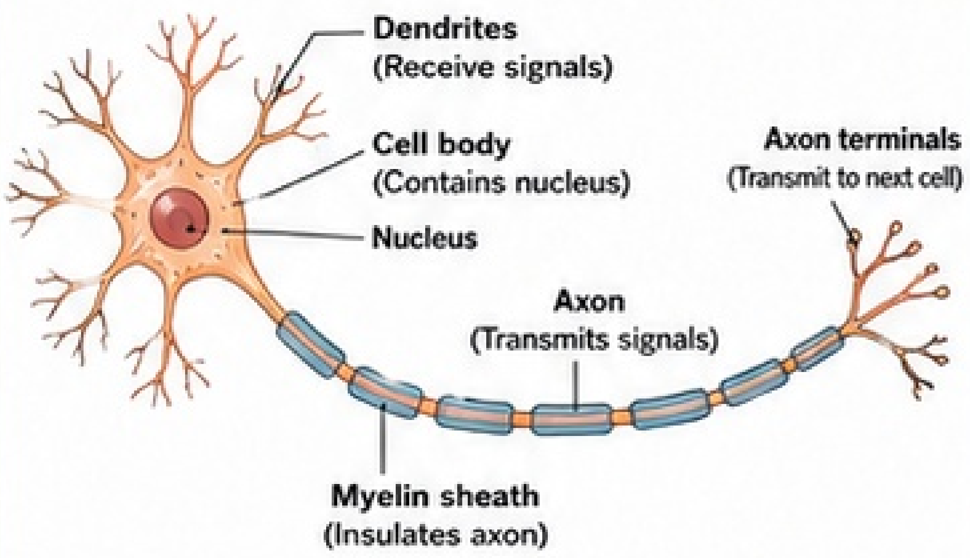
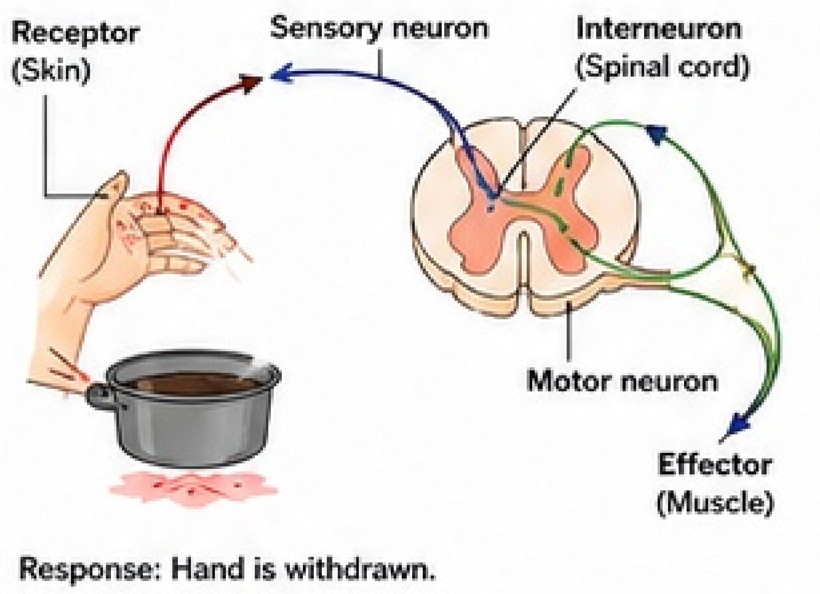
SECTION H - DIAGRAM BANK, DAILY-LIFE EXAMPLES, IMPORTANT OBSERVATIONS AND APPLICATIONS

1 Diagram Bank

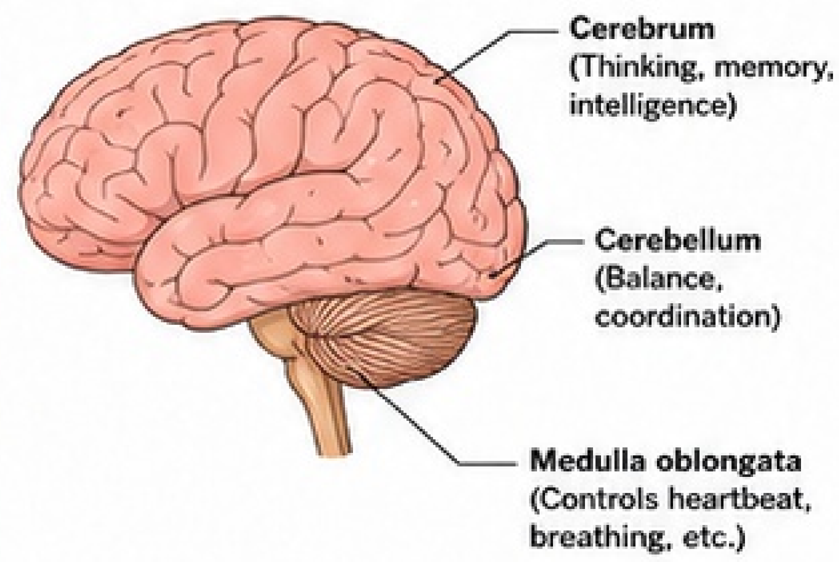
(a) Neuron (Nerve Cell)



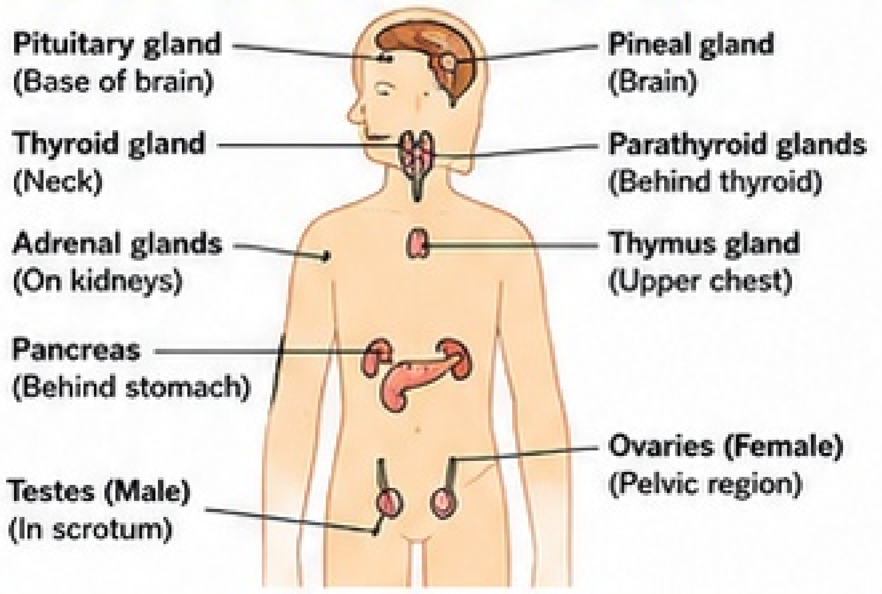
(b) Reflex Arc (e.g., withdrawal of hand from hot object)



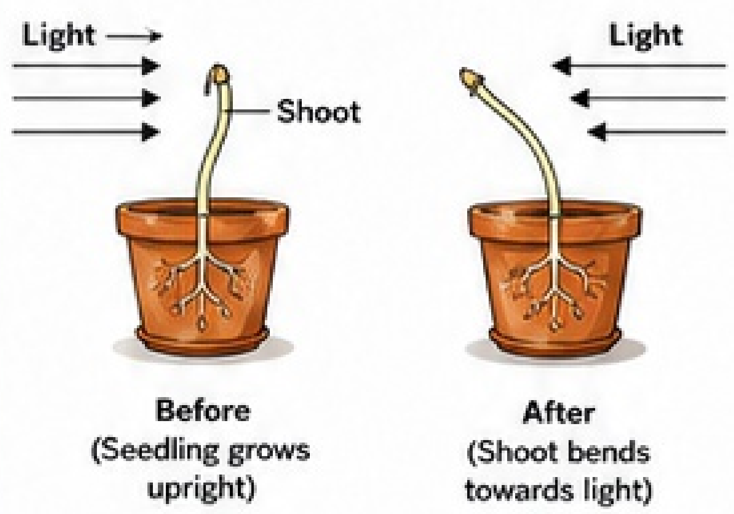
(c) Human Brain (Lateral View)



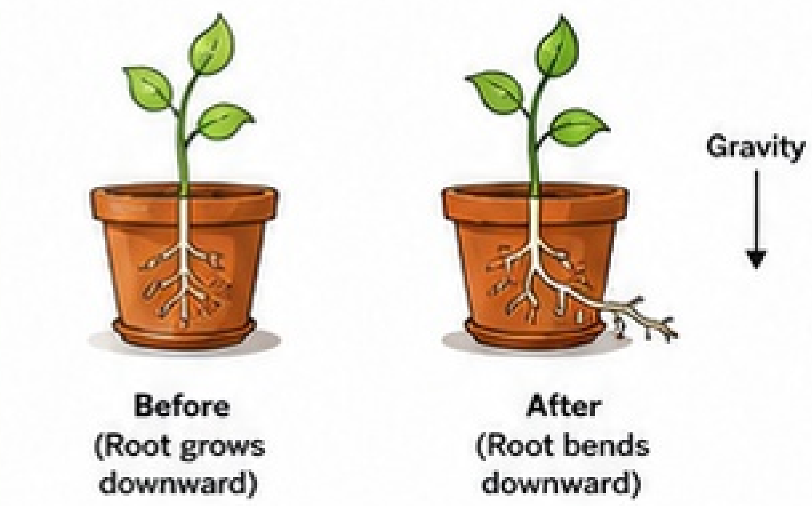
(d) Major Endocrine Glands in Human Body



(e) Phototropism (Shoot towards light)



(f) Geotropism (Root towards gravity)



2 Daily-Life Examples

1. Touching a hot object

Quick withdrawal of hand is a reflex action.

2. Salivation

Sight or smell of food stimulates salivary glands.

3. Balancing while cycling

Cerebellum helps maintain balance and coordination.

4. Adrenaline in fear

Adrenaline released by adrenal glands prepares body to face danger.

5. Plant shoot bending toward light

Shoot shows positive phototropism.

6. Roots growing downward

Roots show positive geotropism.

3 Common Disorders / Imbalances

Disorder	Gland Affected	Cause / Reason	Symptoms	Simple Care / Treatment
Goitre	Thyroid gland	Deficiency of iodine in diet	Swelling in neck, weakness, weight gain, tiredness	Iodised salt, balanced diet; medicines as advised by doctor
Diabetes Mellitus	Pancreas (Islets of Langerhans)	Pancreas does not produce enough insulin or body does not use it properly	Increased thirst, frequent urination, fatigue, weight loss	Insulin injections/ medicines, control sugar intake, exercise
Stunted Growth (Dwarfism)	Pituitary gland	Deficiency of growth hormone during childhood	Height much less than normal for age	Growth hormone therapy (as advised), proper nutrition

4 Important Observations

- Nervous system works very fast through electrical impulses.
- Endocrine system works slowly through hormones in blood.
- Both systems work together to maintain coordination in the body.
- Reflex actions are quick, automatic and protect the body.
- Plants also respond to stimuli (light, gravity, water) for survival.
- Balanced diet, exercise and rest keep the body systems healthy.

5 Applications

- ✓ Reflex actions protect us from danger.
- ✓ Hormones help in growth, development and reproduction.
- ✓ Coordination helps in learning new skills (writing, cycling, sports).
- ✓ Understanding plant responses helps in agriculture and gardening.
- ✓ Hormone medicines are used to treat disorders like diabetes, thyroid problems, growth hormone deficiency.

Exam Tips

- ✓ Draw diagrams with labels and neat lines.
- ✓ For reflex arc, show correct pathway.
- ✓ In brain diagram, label all three parts.
- ✓ For hormones, write names of glands and their main functions.
- ✓ Use key words and short points in answers.
- ✓ Make flowcharts/steps to score more.

Centum Focus Link concepts with real life: Think how your body and plants respond to everyday situations!

Ch 6: Control and Coordination

SECTION I - QUESTION-READY NOTES, COMMON MISTAKES, NCERT TRAPS AND MUST-REMEMBER POINTS

1 One-Mark Question Ready Points

- Control and coordination in multicellular organisms is achieved by nervous system and endocrine system.
- The basic unit of nervous system is neuron.
- Dendrites receive signals; axon carries signals away.
- Synapse is the junction between two neurons.
- Reflex action is a rapid, automatic and involuntary response.
- Spinal cord is the main centre for reflex actions.
- Hormones are chemical messengers.
- Pituitary gland is called the 'master gland'.
- Thyroxine hormone is released by thyroid gland.
- Iodine is essential for synthesis of thyroxine.
- Adrenaline is released by adrenal glands.
- Tropic movements in plants are in response to growth hormones.
- Phototropism is movement of plant towards light.
- Geotropism is movement of plant in response to gravity.

2 Short-Answer Question Ready Points

- **Neuron:** Structural and functional unit of nervous system.
- **Reflex arc pathway:** Receptor → Sensory neuron → Spinal cord / Interneuron → Motor neuron → Effector.
- **Difference:** Nervous system works fast and short-lived; Endocrine system works slow and long-lasting.
- **Endocrine glands:** Pituitary, Thyroid, Parathyroid, Adrenal, Pancreas, Gonads (Testes & Ovaries).
- **Growth hormone:** Secreted by pituitary; helps in growth of bones and muscles.
- **Insulin and Glucagon:** Secreted by pancreas; insulin lowers blood sugar, glucagon raises it.
- **Plant hormones:** Auxins (growth, phototropism), Gibberellins (stem elongation, seed germination), Cytokinins (cell division), Ethylene (ripening of fruits).
- Coordination is necessary for survival and proper functioning of body.

3 Important Definitions

Term	Definition
Neuron	A specialized cell that transmits nerve impulses.
Reflex Action	A quick, automatic and involuntary response to a stimulus.
Hormone	A chemical substance secreted by endocrine glands that regulates body functions.
Endocrine Gland	A ductless gland that releases hormones directly into the blood.
Tropic Movement	Directional growth movement of a plant part in response to an external stimulus.
Phototropism	Growth movement of plant towards or away from light.
Geotropism	Growth movement of plant in response to gravity.

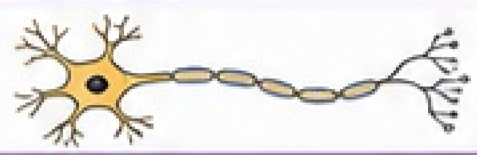
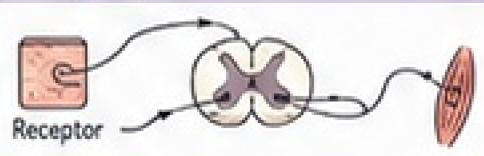
4 Why? (Conceptual Understanding)

	Why is reflex action quick?	Because it does not involve the brain. The impulse travels through a short pathway in the spinal cord called reflex arc.
	Why do plants need hormones?	Hormones regulate growth, development and responses to stimuli like light, gravity and touch.
I 53	**Why is iodine essential for our body?**	*Iodine is needed for the production of thyroxine hormone in the thyroid gland, which regulates metabolism.*
	Why is adrenaline called the 'emergency hormone'?	*Because it prepares the body to face sudden stress or danger by increasing heart rate, breathing rate and energy supply instantly.*

5 Common Mistakes Students Make

Mistake	Correct Understanding
✗ Thinking spinal cord is part of brain.	✓ Spinal cord is part of the central nervous system but not of the brain.
✗ Believing all actions are controlled by brain.	✓ Reflex actions are controlled by spinal cord and do not involve brain for quick response.
✗ Confusing nerves and neurons.	✓ Neuron is a cell; nerves are bundles of axons of many neurons.
✗ *Thinking hormones travel through nerves.*	✓ *Hormones travel through blood, not through nerves.*
✗ *Believing plant movements are due to muscles.*	✓ *Plants lack muscles; movements are due to growth (tropic movements) or turgor changes (nastic movements).*

6 High-Value Labeled Diagrams to Know

Diagram	What to Label	
	**Neuron**	Dendrites, Cell body, Nucleus, Axon, Axon terminals
	**Reflex Arc**	*Receptor, Sensory neuron, Spinal cord (interneuron), Motor neuron, Effector (muscle)*
	**Endocrine Glands**	*Pituitary, Thyroid, Pancreas, Adrenal, Gonads (Testes/Ovaries)*
	**Phototropism in Plant**	*Light source, Shoot bending towards light, Roots, Direction of growth*

7 NCERT Traps – Don't Get Tricked!

Trap Statement (As in Exam)	Reality (Correct Concept)
? *"The brain is responsible for all types of actions."*	**False.** Reflex actions are controlled by spinal cord.
? *"Hormones are secreted by all glands in the body."*	**False.** *Only endocrine glands (ductless glands) secrete hormones.*
? *"Iodine is required for the production of insulin."*	**False.** *Iodine is required for thyroxine, not insulin.*
? *"Plants have nerves and muscles."*	**False.** *Plants have hormones; movements are due to growth or turgor.*

8 Top 10 Must-Remember Points

1. Control and coordination in animals is through nervous and endocrine systems.
2. Neuron is the basic structural and functional unit of nervous system.
3. Reflex actions are quick, automatic and help in protection.
4. Hormones are chemical messengers transported by blood.
5. Pituitary gland is the master gland of the body.
6. Thyroxine regulates *metabolism; iodine is essential for its synthesis.*
7. Insulin *lowers blood sugar; glucagon raises blood sugar.*
8. Adrenaline *prepares the body for fight or flight situations.*
9. Plant hormones regulate growth and directional movements.
10. Coordination *ensures proper functioning and survival of organisms.*

Ch 6: Control and Coordination

SECTION J - MASTER REVISION SHEET

1 CHAPTER SUMMARY (At a Glance)

- Living organisms sense changes inside and outside their body.
- Control and coordination help in quick and appropriate responses.
- In animals, control is by nervous system; coordination is by hormones.
- Nervous system works fast and for short duration.
- Endocrine (hormonal) system works slow and for long duration.
- Plants also respond to stimuli through growth movements (tropisms).
- Both systems together help maintain balance and survival.

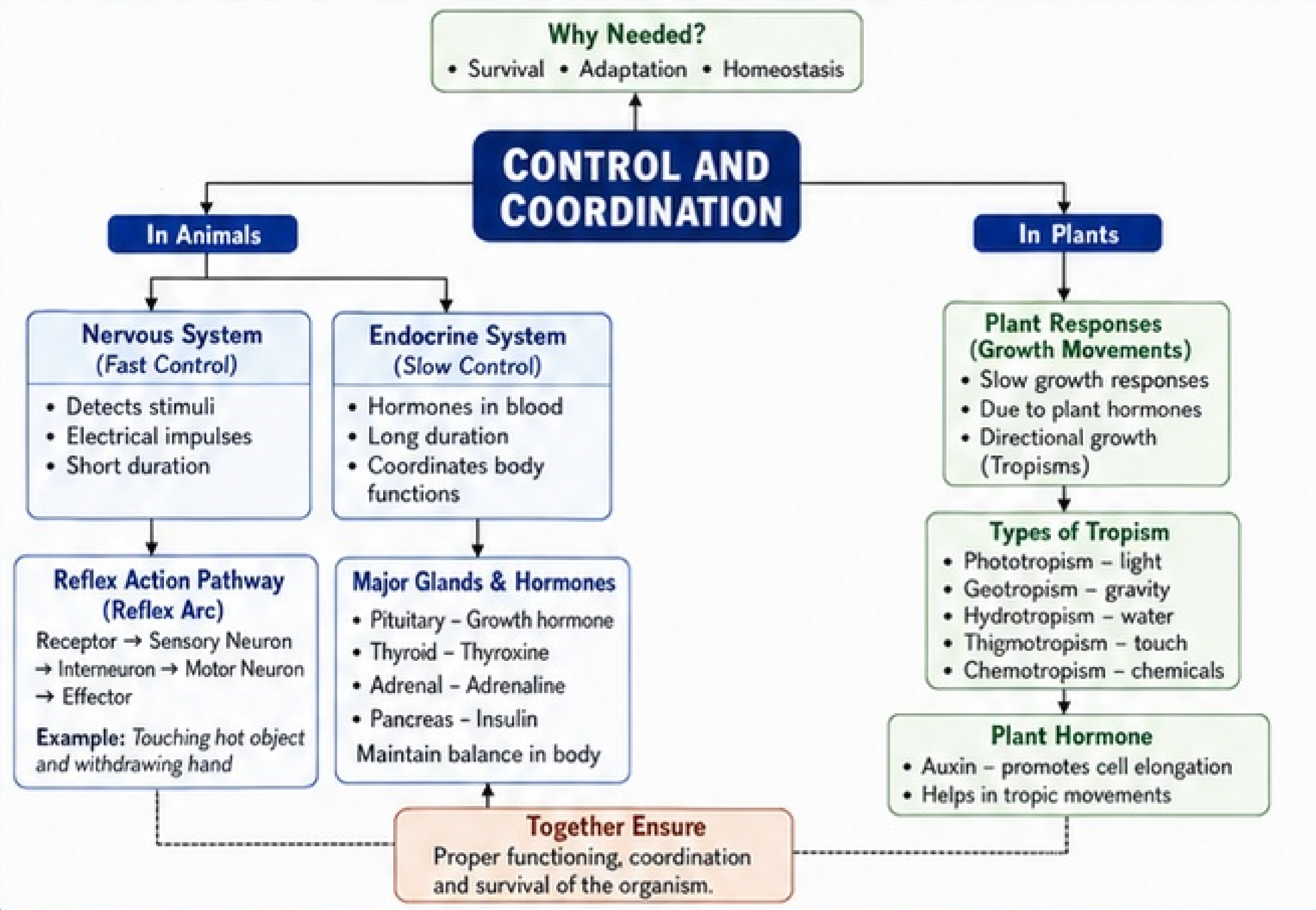
2 KEY DEFINITIONS

Stimulus	Any change in the internal or external environment that causes a response.
Response	The action or reaction shown by an organism to a stimulus.
Receptor	A specialized structure that detects a stimulus and sends the message.
Effector	A muscle or gland that brings about the response.
Neuron	A nerve cell that carries electrical impulses.
Synapse	The junction between two neurons where the impulse passes.
Reflex Action	A quick, automatic and involuntary response to a stimulus.
Reflex Arc	The pathway taken by impulse during a reflex action. (Receptor → Sensory Neuron → Interneuron → Motor Neuron → Effector)
Hormone	A chemical messenger secreted by endocrine glands into blood to act on target organs.
Tropism	Directional growth movement of a plant in response to a stimulus.
Auxin	A plant hormone that promotes cell elongation and is responsible for tropic movements.
Adrenaline	A hormone secreted by adrenal glands during stress or emergency; increases heart rate, breathing and energy supply.

3 FORMULATED ANSWER FRAMES

1. **What is control and coordination?**
   Ans: Control and coordination help organisms to sense changes and respond appropriately to survive in their environment.
2. **Differentiate between nervous and hormonal control.**
   Ans: Nervous control is fast, short-term and through electrical impulses. Hormonal control is slow, long-term and through chemicals (hormones).
3. **What is a reflex action? Give an example.**
   Ans: *A reflex action is a quick, automatic and involuntary response.*
   Example: *Withdrawal of hand on touching a hot object.*
4. **Draw and label the reflex arc.**
   Ans: Receptor → Sensory Neuron → Interneuron → Motor Neuron → Effector
5. **Name the major gland that secretes adrenaline. State its function.**
   Ans: Adrectal glands secrete adrenaline. It prepares the body for emergencies by increasing heartbeat, breathing rate and energy.
6. **What is tropism? Give two examples.**
   Ans: *Tropism is growth movement of plants in response to a stimulus.*
   Examples: *Phototropism (towards light), Geotropism/Gravitropism (roots downward, shoots upward).*
7. **Why are plants said to have slow responses?**
   Ans: *Plants do not have nervous system; responses occur through growth hormones and are slower than in animals.*

4 CONCEPT MAP – CONTROL AND COORDINATION



5 CENTUM CHECKLIST – WHAT TO REVISE

- [ ] Understand control vs coordination.
- [ ] Nervous system: structure and function of neuron.
- [ ] Reflex action: meaning, reflex arc, example.
- [ ] Difference between voluntary and involuntary actions.
- [ ] Endocrine glands and their hormones & functions.
- [ ] Comparison: nervous vs endocrine control.
- [ ] Plant responses: tropisms and their types.
- [ ] Role of auxin in plant growth movements.
- [ ] Examples from daily life and applications.
- [ ] Diagrams: neuron, reflex arc, tropisms.

6 BEFORE EXAM – QUICK GLANCE TABLE

Topic	Key Points	Examples / Facts	Remember
Nervous System	Fast, electrical impulses, through neurons	Brain, spinal cord, nerves	Quick but short-lived responses
Reflex Action	Automatic, involuntary, protective	Pulling hand from hot object	Reflex Arc: Receptor → Sensory → Interneuron → Motor → Effector
Endocrine System	Slow, chemical (hormones), long-lasting	Pituitary, Thyroid, Adrenal, Pancreas	Hormones travel in blood to target organs
Adrenaline	*Emergency hormone*	*Adrenal glands*	Increases heart rate, breathing, energy
Plant Responses	*Growth based, slow*	*Tropisms*	*Due to plant hormones*
Auxin	*Plant hormone*	*Promotes cell elongation*	*Responsible for tropic movements*

7 ONE-LINE TIPS (Remember Fast!)

- ✓ Nervous system = Fast & Electrical.
- ✓ Endocrine system = Slow & Chemical.
- ✓ Reflex actions = Quick, automatic, protective.
- ✓ Hormones = Chemical messengers in blood.
- ✓ Plants respond through growth, not movement.
- ✓ Auxin = Key hormone for plant tropisms.

8 EXAM STRATEGY

- ★ Draw neat and labeled diagrams.
- ★ Use flowcharts/tables for comparisons.
- ★ Write definitions in one crisp line first.
- ★ Give examples to strengthen answers.
- ★ Answer in steps: Point → Explain → Example.
- ★ Underline key terms in your answer.

CENTUM FOCUS

To score full marks:

- ✓ Learn every definition word-to-word.
- ✓ Practice all diagrams till perfect.
- ✓ Revise tables and differences repeatedly.
- ✓ Solve previous year questions.

Be Concept Clear. Be Confident. Score Centum!